

FILE NAME: Y:\COCONUS\MultipleSheets\2024_1783264-P1360_LAMFA-B_Design\DWG\Sheet\04_GEO\1783264-B-001.dwg LAYOUT NAME: B-001 PLOTTED: Wednesday, January 21, 2026 - 8:07am USER: bropatton

D

C

B

A

Michael Baker INTERNATIONAL			DEFINITION OF SOIL TERMS (Page 1 of 3)				
Soil Description (ASTM D2488)			Grain Size (ASTM D2488)				
Criteria	Group Symbol	Primary Group Name	Material	Size (mm)	Size (in)	US sieve	
< 5% Passing #200 sieve	< 5% passing #200 sieve	GP	Poorly graded gravel	Boulder	> 300	> 12	
		GW	Well-graded gravel	Cobble	75 to 300	3 to 12	
	> 15% passing #200 sieve	GM	Silty gravel	Gravel			
		GC	Clayey gravel	Coarse	19 to 75	0.75 to 3	¾ to 3
	5-15% passing #200 sieve	GW-GM	Well-graded gravel with silt	Fine	4.75 to 19	0.187 to 0.75	#4 to ¾
		GW-GC	Well-graded gravel with clay				
		GP-GM	Poorly graded gravel with silt				
		GP-GC	Poorly graded gravel with clay				
	< 5% passing #200 sieve	SP	Poorly graded sand				
		SW	Well-graded sand				
% gravel < % sand	> 15% passing #200 sieve	SM	Silty sand				
		SC	Clayey sand				
	5-15% passing #200 sieve	SW-SM	Well-graded sand with silt				
		SW-SC	Well-graded sand with clay				
		SP-SM	Poorly graded sand with silt				
		SP-SC	Poorly graded sand with clay				
	> 50% passing #200 sieve	ML	Silt				
		MH	Elastic Silt				
		OL	Organic Silt				
		CL	Lean Clay				
		CH	Fat Clay				
		OH	Organic Clay				
			Angularity (ASTM D2488)				
			Angular	Particles have sharp edges and relatively plane sides with unpolished surfaces			
			Subangular	Particles are similar to angular description but have rounded edges			
			Subrounded	Particles have nearly plane sides but have well rounded corners and edges			
			Rounded	Particles have smoothly curved sides and no edges			
			Particle Shape (ASTM D2488)				
			Flat	Particles with width/thickness >3.			
			Elongated	Particles with length/width >3.			
			Flat and elongated	Particles meet criteria for both flat and elongated			
			Particle Hardness (ASTM D2488)				
			particles are hard (particles do not fracture under considerable hammer blow)				
			(particles fracture under considerable hammer blow)				
			(particles crush under hammer blow)				
			(particles crush under hand pressure)				
NOTES: (ASTM D2488) "Standard Practice for Description and Identification of Soils (Visual-Manual Procedure) ", American Society of Testing and Materials, Standard D-2488, 2017							

Michael Baker
INTERNATIONAL

DEFINITION OF SOIL TERMS (Page 2 of 3)

Continued from previous page

Plasticity of Fines (ASTM D2488)

Non Plastic	A 1/8 inch (3 mm) thread cannot be rolled at any water content.
low plasticity	The thread can barely be rolled, and the lump cannot be formed when drier than the plastic limit.
medium plasticity	The thread is easy to roll, and not much time is required to reach the plastic limit. The thread cannot be rerolled after reaching the plastic limit. The lump crumbles when drier than the plastic limit.
high plasticity	It takes considerable time rolling and kneading to reach the plastic limit. The thread can be rolled several times after reaching the plastic limit. The lump can be formed without crumbling when drier than the plastic limit.

Dry Strength (ASTM D2488)

no dry strength	The dry specimen crumbles with mere pressure of handling.
low dry strength	The dry specimen crumbles with some finger pressure.
medium dry strength	The dry specimen breaks into pieces or crumbles with considerable finger pressure.
high dry strength	The dry specimen cannot be broken with finger pressure. Specimen will break into pieces between thumb and a hard surface.
very high dry strength	The dry specimen cannot be broken between thumb and a hard surface.

Dilatancy (ASTM D2488)

no dilatancy	No visible change in the specimen.
slow dilatancy	Water slowly appears on the surface of the specimen during shaking and does not disappear or disappears slowly upon squeezing.
rapid dilatancy	Water quickly appears on the surface of the specimen during shaking and disappears upon squeezing.

Toughness (ASTM D2488)

low toughness	Only slight pressure is required to roll the thread near the plastic limit. The thread and the lump are weak and soft.
medium toughness	Medium pressure is required to roll the thread to near the plastic limit. The thread and the lump have medium stiffness.
high toughness	Considerable pressure is required to roll the thread to near the plastic limit. The thread and the lump have very high stiffness.

Reaction to HCL (ASTM D2488)

no reaction to HCl	No visible reaction.
weak reaction to HCl	Some reaction, with bubbles forming slowly.
strong reaction to HCl	Violent reaction, with bubbles forming immediately.

Moisture Condition (ASTM D2488)

Dry	Absence of moisture, dusty, dry to the touch
Moist	Damp but no visible water
Wet	Visible free water, usually the soil is below the water table

Relative Moisture

NP	A 1/8 inch (3 mm) thread cannot be rolled at any water content
-PL	Breaks up at larger than a 1/8 inch (3 mm) diameter roll of soil
+PL	Cohesive to a 1/8 inch (3 mm) diameter roll of soil
+LL	Conforms to container when shaken

NOTES: (ASTM D2488) "Standard Practice for Description and Identification of Soils (Visual-Manual Procedure) ", American Society of Testing and Materials, Standard D-2488, 2017

TEMPLATE-1050, PROJECT, STRATA, SYMBOLS PROJECT-MOL POINT SMG PRG GP, LIBRARY=BAKER, 10-1-GLB, DATE PRINTED: 22724, S.O.#=19789

Michael Baker INTERNATIONAL			STRATIGRAPHIC SYMBOLS		
	Well graded Gravel (GW)		Poorly-graded Gravel (GP)		Silty Gravel (GM)
	Well graded Sand (SW)		Poorly-graded Sand (SP)		Silty Sand (SM)
	Lean Clay (CL)		Topsoil		Asphalt
					Clayey Gravel (GC)
					Clayey Sand (SC)
					Not Described
Miscellaneous Symbols					

TEMPLATE-1050, PROJECT, STRATA, SYMBOLS PROJECT-MOL POINT SMG PRG GP, LIBRARY=BAKER, 10-1-GLB, DATE PRINTED: 22724, S.O.#=19789

Michael Baker INTERNATIONAL		DEFINITION OF SOIL TERMS (Page 3 of 3)	
In-place conditions			
Apparent Density of Granular Soils (FHWA NHI-06-088)			
Term	SPT N60 (BPF or B/0.3 m)	Response to ½" (6 mm) rod driven with 5lb (2.3 kgf) hammer**	Relative Density (%)
Very Loose	0 - 4		
Loose	5 - 10	Easily penetrated by hand	0-40
Medium Dense	11 - 30	Easily penetrated with hammer	40-70
Dense	31 - 50	Penetrates 1 foot (0.3 m)	70-85
Very Dense	> 50	Penetrates 1 inch (2.5 cm)	85-100
Consistency of Cohesive Soils (FHWA NHI-06-088)			
Term	SPT N60 (BPF or B/0.3 m)	Unconf. comp. strength (TSF or Kg/cm²)	Results of Manual Manipulation
Very soft	< 2	< 0.25	Specimen (height = twice diameter) sags under its own weight, extrudes between fingers when squeezed
Soft	2 - 4	0.25 - 0.5	Specimen can be pinched in two between thumb and forefinger, remolded by light finger pressure
Medium stiff	4 - 8	0.5 - 1	Imprinted easily with fingers, remolded by strong finger pressure
Stiff	8 - 15	1 - 2	Imprinted by considerable finger pressure or indented by thumbnail
Very stiff	15 - 30	2 - 4	Barely imprinted by finger or indented by thumbnail
Hard	> 30	> 4	Cannot be imprinted by finger, difficult to indent by thumbnail
Fabric/Structure (ASTM D2488)			
Stratified	Alternating layers of varying material or color with layers at least ¼ inch (6 mm) thick.		
Laminated	Alternating layers of varying material or color with the layers less than ¼ inch (6 mm) thick.		
Fissured	Breaks along definite planes of fracture with little resistance to fracturing		
Stickensided	Fracture planes appear polished or glossy, sometimes striated.		
Blocky	Cohesive soil that can be broken down into small angular lumps that resist further breakdown		
Lensed	Inclusion of small pockets of different soils, such as small lenses of sand scattered through a mass of clay		
Homogeneous	Same color and appearance throughout		
Heterogeneous*	Contains many distinctly different materials randomly arranged (as in an embankment fill)		
Cementation (ASTM D2488)			
weak cementation	Crumbles or breaks with handling or little finger pressure.		
moderate cementation	Crumbles or breaks with considerable finger pressure.		
strong cementation	Will not crumble or break with finger pressure.		
NOTES: (ASTM D2488) "Standard Practice for Description and Identification of Soils (Visual-Manual Procedure) ", American Society of Testing and Materials, Standard D-2488, 2017			
(FHWA NHI-06-088) "Soils and Foundation Workshop – Reference Manual", U.S. Department of Transportation, Federal Highway Administration, Publication No. FHWA NHI-06-088, December 2006			
* Not part of ASTM standard D2488, added for completeness.			
** Rod test behavior added from NHI Publication FHWA HI-97-021 "Subsurface Investigations", November 1997.			

TEMPLATE-1050, SOIL, LEGEND-PAGE003, PROJECT-MOL POINT SMG PRG GP, LIBRARY=BAKER, 10-1-GLB, DATE PRINTED: 22724, S.O.#=19789

NOTES

- THE SOIL BORING RECORDS REPRESENT THE GEOTECHNICAL ENGINEER'S INTERPRETATION OF THE SUBSURFACE CONDITIONS ENCOUNTERED AT THE SPECIFIC BORING AND TEST PIT LOCATIONS AT THE TIME OF THE FIELD EXPLORATION BASED ON VISUAL EXAMINATION OF THE FIELD SAMPLES OBTAINED AND LABORATORY CLASSIFICATION TESTING ON SELECTED SAMPLES.
- LINES DESIGNATING THE INTERFACE BETWEEN VATIOS STRATA ON THE BORING RECORDS REPRESENT THE APPROXIMATE INTERFACE LOCATION. THE ACTUAL TRANSITION BETWEEN STRATA MAY BE MORE GRADUAL THAN INDICATED. THE BORING RECORDS MAY NOT NECESSARILY BE REPRESENTATIVE OF THE MATERIAL CHARACTER AND STRATA THICKNESS OVER THE ENTIRE CONSTRUCTION AREA.
- WATER LEVELS SHOWN ON THE BORING RECORDS REPRESENT THE CONDITIONS ENCOUNTERED AT THE TIME OF THE FIELD EXPLORATION AND DO NOT NECESSARILY REPRESENT GROUNDWATER LEVELS THAT WILL BE ENCOUNTERED DURING CONSTRUCTION. IT IS POSSIBLE THAT GROUNDWATER LEVELS WILL FLUCTUATE DUE TO SEASONAL VARIATIONS AND OTHER FACTORS.
- TESTING LOCATIONS AND ELEVATIONS SHALL BE CONSIDERED APPROXIMATE.
- BORINGS WERE ADVANCED USING MUD ROTARY DRILLING METHODS AS NOTED.
- BORINGS PERFORMED BY AECOM-BAKER AND REPORTED IN AECOM-BAKER'S SUBSURFACE INVESTIGATION DATA REPORT DATED 29 MARCH 2024 ARE SHOWN.
- REFER TO BORING LOG LEGEND ON THIS SHET FOR AN EXPLANATION OF THE SYMBOLS USED ON BORING RECORDS.



APPROVED		A/E INFO
FOR COMMANDER NAVFAC		
ACTIVITY		
D. EVANS VIA EMAIL		
SATISFACTORY TO	DATE	01/15/2026
DES LAP	DRW LAP	CHK CFC
PM/DM	MJD/ATP	
BRANCH MANAGER		
DES PROD DIR	WBF	
FIRE PROTECTION ENGINEER	DSN	
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC		
NAVFAC DRAWING NO. 14157070		
SHEET 8 OF 170		
B-001		

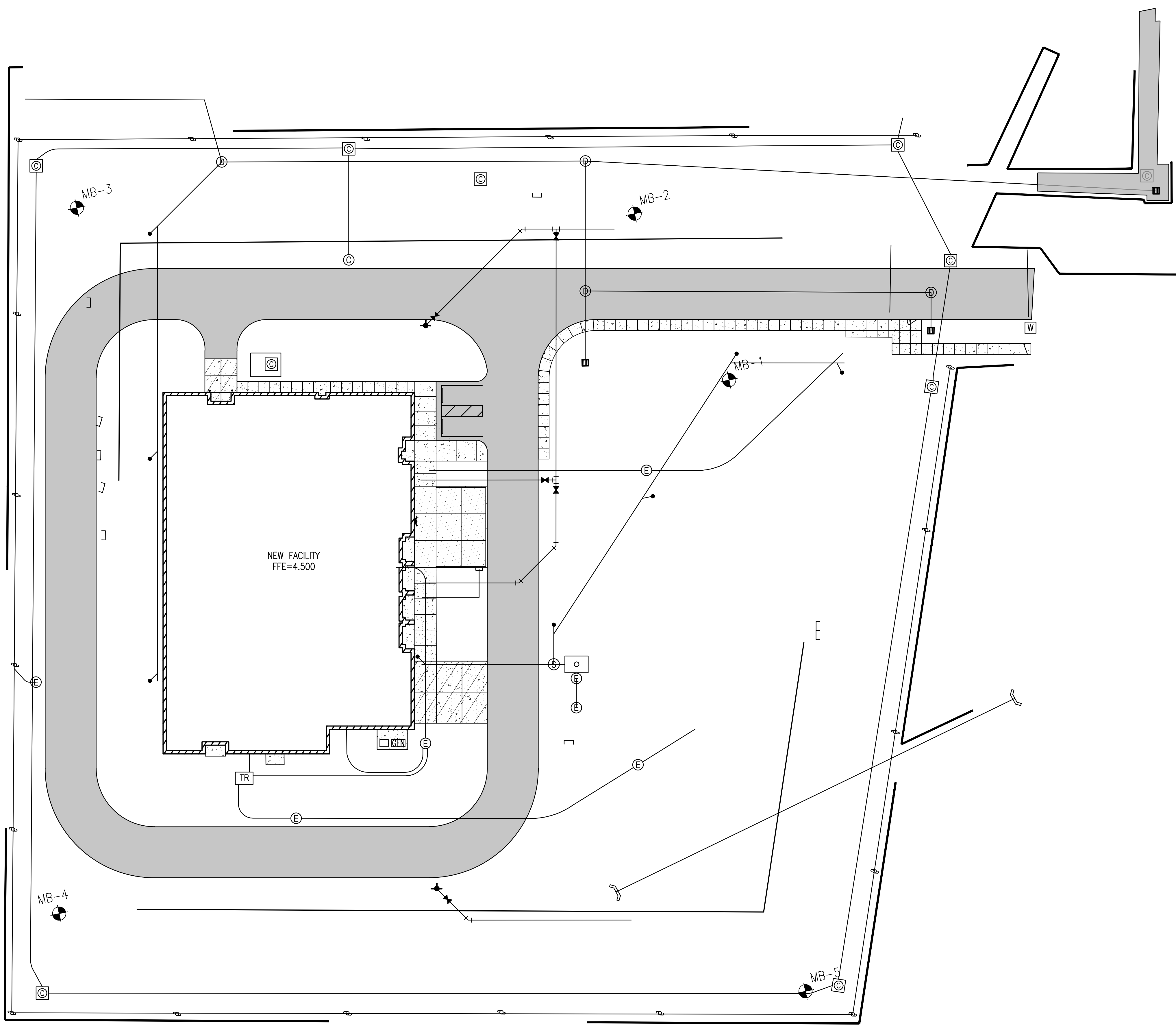
FILE NAME: Y:\OCONUS\Multiplatform\2026_1783264_P1360_JMFB\B_Design\DWG\Sheet\04_GEO\1360-1783264-B-100.dwg LAYOUT NAME: B-101 PLOTTED: Wednesday, January 21, 2026 - 8:07am USER: brampton

D

C

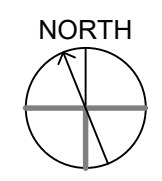
B

A



LEGEND

MB-1
BORING LOCATION



0 7.5 15m

DEPARTMENT OF THE NAVY		NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND		NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - ATLANTIC		NORFOLK, VA	
PLANNING, DESIGN AND CONSTRUCTION		VARIOUS LOCATIONS		VARIOUS LOCATIONS		VARIOUS LOCATIONS	
JOINT MARITIME FACILITY		SUBSURFACE EXPLORATION LOCATION PLAN					
SCALE: AS NOTED		EPROOFED NO.: 1838720		NAVFAC DRAWING NO. 14157071		SHEET 9 OF 170	
CONSTR. CONTR. NO.				B-101			
APPROVED		FOR COMMANDER NAVFAC		ACTIVITY		D. EVANS VIA EMAIL	
SATISFACTORY TO		DATE		01/15/2026			
DESIGN		LAP		DRAW		LAP	
PM/DM		MJD/ATP		CFC			
BRANCH MANAGER		WBF		DSN			
DESIGN PROD. DIR.							
FIRE PROTECTION ENGINEER							
APPROVED		A/E INFO		SEAL		DATE	
SYN		DESCRIPTION		DATE		APPR	

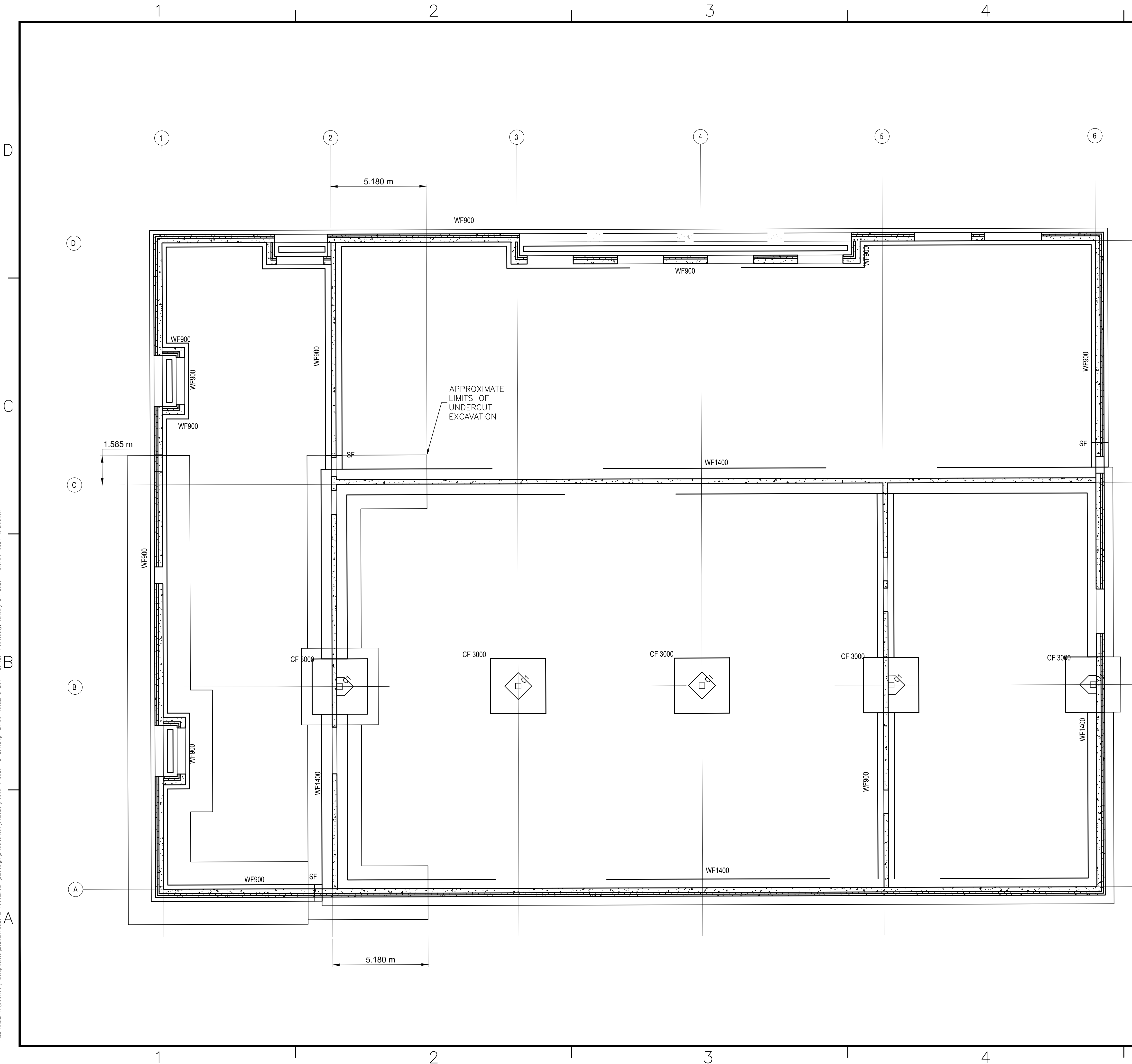


D

C

B

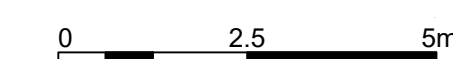
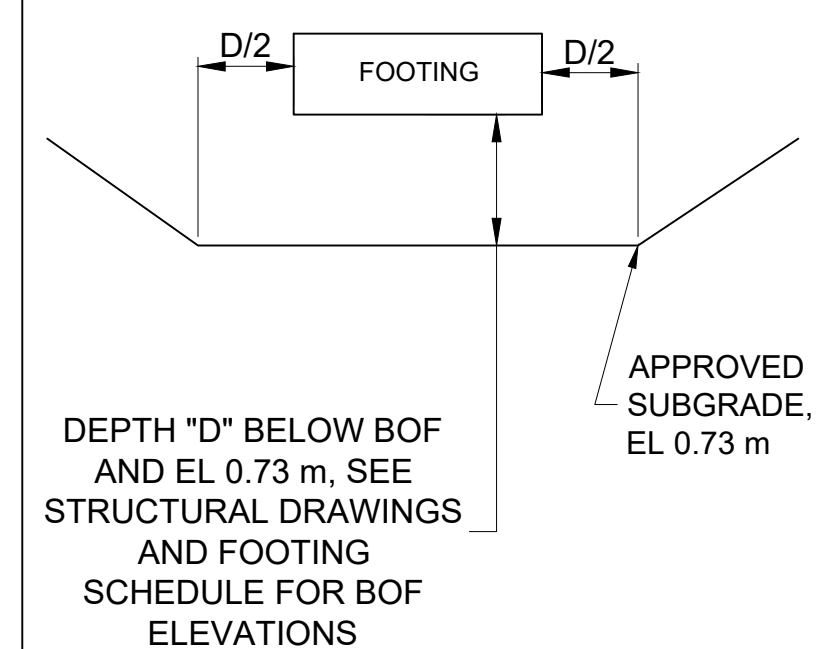
A



NOTES

1. PERFORM UNDERCUT EXCAVATION TO MINIMUM ELEVATION 0.73 m BENEATH FOOTINGS WITHIN THE AREAS IDENTIFIED ON THIS PLAN. THIS INCLUDES FOOTINGS ALONG COLUMN LINES 1 AND 2 FROM COLUMN LINES A THROUGH C AND FOOTINGS ALONG COLUMN LINES A AND C BETWEEN COLUMN LINES 1 AND 2.
2. FOR UNDERCUT DEPTH "D" THE UNDERCUT EXTENTS MUST BE EXTENDED "D/2" WIDTH TO EITHER SIDE OF THE FOOTING. SEE UNDERCUT DETAIL.
3. THE CONTRACTOR MAY CONDUCT TEST PITS TO THE PLANNED UNDERCUT ELEVATION TO CONFIRM THE PRESENCE OF THE TIDAL MARSH DEPOSIT IDENTIFIED IN MB-3. IN THE EVENT THAT THE TEST PITS INDICATE THIS LAYER IS NOT PRESENT WITHIN THE FOOTING UNDERCUT EXCAVATION LIMITS, THEN THE CONTRACTOR MAY PROPOSE A COST CREDIT VARIANCE TO ELIMINATE THE UNDERCUT FROM THE PROJECT.
4. CONTRACTOR IS RESPONSIBLE FOR DEWATERING OF UNDERCUT EXCAVATIONS AND MAINTAINING STABLE UNDERCUT EXCAVATION SIDES USING VERTICAL SHORING AND/OR SAFE SLOPES AND SAFE WORKING CONDITIONS.
5. SEE 31 00 00 EARTHWORK FOR BACKFILL MATERIAL AND COMPACTION REQUIREMENTS.

UNDERCUT DETAIL, NTS

[illegible]